

SMART INDIA HACKATHON PROJECT SUBMISSION

Democratizing Sports Talent Assessment

An AI-Powered Mobile Platform for Grassroots Identification designed to bring professional sports lab analytics to any village in India using a standard smartphone.



Team Arjuna



Mobile Grassroots Identification

| The Challenge



Geographic Barrier

Talent in rural and remote communities remains completely undiscovered due to a lack of professional testing infrastructure and sports academy access.



High Scouting Costs

Physical scouting camps and multi-camera sports labs are highly expensive to organize, impossible to scale nationally, and logistically restricted.



Human Bias

Subjective evaluation metrics and physical favoritism by scouts can lead to severe inconsistency, rendering selection parameters flawed and unfair.

| The Solution

Talent Track Mobile App

An AI-first mobile platform that enables any athlete with a smartphone to self-record athletic and technical trials.

-  **Smartphone Delivery:** Uses a single RGB standard phone camera. No external hardware required.
-  **On-Device CV:** Computes complex performance variables locally on consumer smartphones.
-  **Data Security:** Decentralized verification and absolute data integrity to prevent manipulation.



| AI Powered Analysis



Pose Estimation

Real-time edge skeletal tracking using MediaPipe API models to detect 33 human joint coordinates dynamically.



Metric Extraction

Automated cloud/edge computations to extract vertical jump heights, precise limb flight times, and angle speeds.



Cheat Detection

Lightweight validation layer checking frame ratios, tracking anomalies, and visual deepfakes to filter out fraudulent trials.



Offline Processing

Optimized compiled TensorFlow Lite architectures allowing local feature extractions without real-time internet.

UML Use Case Diagram



ATHLETE

PRIMARY ACTOR



Create Profile



Record & Upload



View Performance



Compete Leaderboard



Shared Use Case: Secure Authentication



SCOUT & ADMIN

SECONDARY AUTHORITIES



Search Metrics



Compare Athlete



Validate AI Flags



Manage Standards



Shared Use Case: Video Verification

| System Data Pipeline Flow

STAGE 01

Record

Local video buffer captures trials with automated sensor angle framing. Compiles offline MP4 chunks.

STAGE 02

Edge AI

MediaPipe tracks 33 skeleton coordinates inside TFLite. Flags velocity anomalies instantly.

STAGE 03

Cloud

S3 video uploads triggered. FastAPI executes deeper, robust anti-cheat & score compiles.

STAGE 04

Analytics

Updates active geographic leaderboard maps. Direct digital scout invitation dispatch.

Entity-Relationship Diagram

ATHLETE 	
PK athlete_id	UUID
name	VARCHAR
age	INT
region_code	VARCHAR
sport_focus	VARCHAR

TRIAL VIDEO 	
PK trial_id	UUID
FK athlete_id	UUID
s3_video_url	VARCHAR
cheat_score	FLOAT
status	ENUM

SHORTLIST 	
PK entry_id	UUID
FK scout_id	UUID
FK athlete_id	UUID
date_added	DATE

AI METRIC 	
PK metric_id	UUID
FK trial_id	UUID
vertical_jump	FLOAT
limb_angle	FLOAT
execution_speed	FLOAT

SCOUT 	
PK scout_id	UUID
agency_name	VARCHAR
assigned_region	VARCHAR

CONCLUSION

Village Grounds to National Glory

Empowering the next generation of athletic champions through precise, objective, and globally accessible AI sports tech.

 Talent Track Assessment Platform

 Team Arjuna | Smart India Hackathon

| Image Sources



<https://www.it-jim.com/wp-content/uploads/2026/02/blog-apps-with-pose-estimation.webp>

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